MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548450

(AMPLIFIER, CONTROL)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the control amplifier used in the Range Computer CP-888/VPS-2, part Of Radar Set AN/VPS-2a

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General specification for "

DRAWINGS

U.S. Army, Frankford Arsenal

F1054845O -	Amplifier, Control

PACKAGING DATA SHEET

10548450	Packaging of Control Amplifier

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(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting *officer.)*

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548450 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - *As* specified in MIL-C-506S2.

3.4 Electrical - Each conductor shall be electrically continuous as specified on the schematic diagram of the control amplifier.

3.5 Performance. - When supplied with the input voltages specified in Table I, and the input signals specified in Table II, the control *amplifier shall* yield the signals at the output terminals and control amplifier assembly test points specified in Table II

TABLE I. - INPUT VOLTAGES

CHARACTERISTICS	TERMINALS
Positive and negative 15.0 ± 0.1 Vdc, regulated; with an ac ripple less than 50 mV peak-to-peak	P1-36, P1-37 (+15 V) P1-22, P1-23, P1-34, P1-35 (± 15 V returns) P1-2O, P1-21 (-15 v)

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TABLE II. - INPUT AND OUTPUT SIGNAL CHARACTERISTICS

INPUTS	OUTPUTS			
FREQUENCY	P1-43 (TP-1)	(TP2)	P1-49 (TP3)	P1-5 (TP4)
5 kHz sine wave	200 mV peak-to-peak	2.0 ± 0.5V peak-to-peak	0.65 ± 10 Vdc	l.0 ± 0.l V peak-to-peak. (See note.)
5 kHz sine wave	1.0V peak-to-peak		3.5 ± O. 5 Vdc with 25 mV ripple peak-to-peak max	5.0 ± 0.5V peak-to-peak
10 kHz sine wave	200 mV peak-to-peak			1.0 v peak-to-peak
500 Hz sine wave	200 mV peak-to-peak			1.0 v peak-to-peak
1 kHz triangular wave	2.25 ● 0. 25 V peak-b-peak		5.5 ± 0.5 Vdc	Output level just at point where wave-forms begin to flatten.

NOTE: With input signal at P1-43 as specified, output at P1-5 shall be adjustable by R13 to this value.

4. QUALITY ASSURANCE PROVISION'S

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

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4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

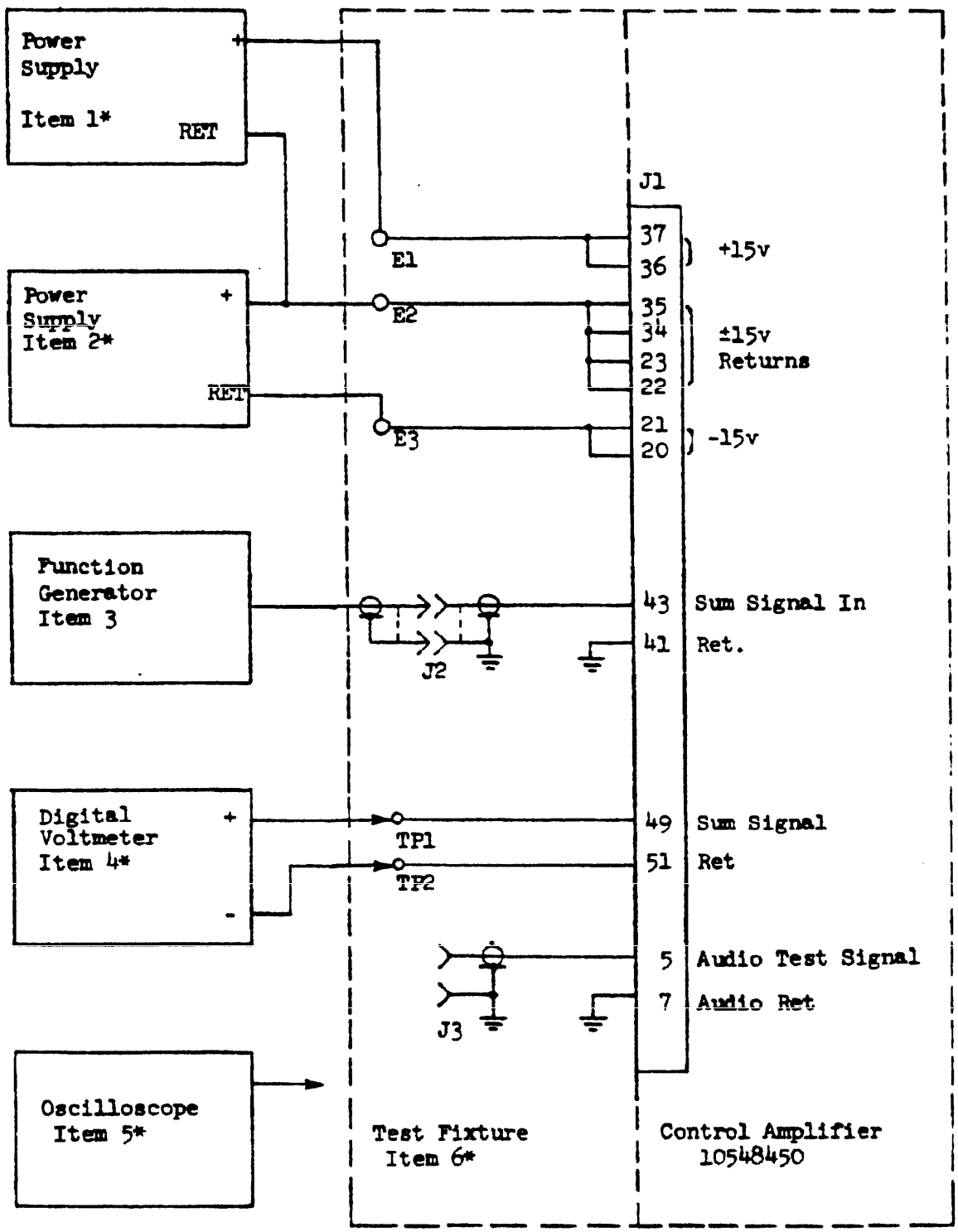
4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652,

4.6,2.3 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table III shall be utilized in the performance of the applicable tests as specified in 4.6.3.

TABLE III. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	15.0 ± 0.1 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 200 mA.
2. Power. supply	Same as item 1.
3. *Function* generator	Sine wave and triangular output waveforms. frequency 500 Hz to 10 kHz. Amplitude 3.0 V peak-to-peak or greater, into 50-ohm load. Sine wave distortion 0. 5% or less.
4. Digital voltmeter	Visual display 3 digits or more, full scale range 20.0 V or greater, Input resistance 1 megohm or greater. *Accuracy* ± 0.01%
5. oscilloscope	Sweep period 100 us/cm or greater, input resistance greater than 8 megohms, input capacitance 25 pF or less.
6. Test fixture	See Figure 1. See Table IV for test fixture parts list.

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*Refer to 4.5.2.3

FIGURE 1. TEST SETUP.

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TABLE IV. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1 thru E2	Binding post, H. H. Smith 257 or equivalent.
J1	Connector; any of the following or equivalent.
	(1) 5893232-1, Malco Mfg. Co., Inc, Chicago, Illinois.
	(2) 500946-01, National Connector Corp. Minneapolis, Minn.
	(3) 07-5422-0198-01, Elco Corp. Willow Grove, Pa.
J2	Connector, receptacle, UG-625/B/ U or equivalent.
J3	Same as J2.
T PI	Tip jack, H. H. Smith 1501 or equivalent.
TP2	Same as TP1.

4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.& 3 Performance. - The selected control amplifiers shall be tested as follows: Set up equipment as shown in Figure 1. Use items 1 and 2 to supply input voltages as specified in Table I. Apply input voltages to E1 through E3 on test fixture, item 6. Use function generator, item 3, to supply input signals as specified in Table 11. Use oscilloscope, item 5, for observation of input signals and output signals as specified in Table II, Use digital voltmeter, item 4, to observe output signal at P1-49 and TP3 on the control amplifier as specified in Table II. Input signal to P1-43 as specified in Table II shall be applied to J2 on test fixture, item 6. Output signals at P1-49 and P1-5 specified in Table II shall be observed at TP1 and J3 respectively, on test fixture, item 6.

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5. PREPARATION FOR DELIVERS

5.1 Packaging, packing and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548450. The level of protection shall be as specified in the procurement document.

6. NOTES

6,1 Intended use. - The control amplifier assembly is used the Range Computer CP-888/VPS-2, part of **Radar Set AN/VPS-2. The control amplifier assembly amplifies and** *converts* **the sum doppler signal into** a dc level equal to the **rms value of the doppler signal. The control amplifier also provides a doppler audio signal for monitoring with a head** *set.*

6.2 **Ordering data. -** Procurement documents should speci the **following:**

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 6.1).

c. Selection of applicable level of preservation packaging: and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

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Preparing Activity:

Army - MU

Project No. 1285-A253

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

NAME *(Print or type name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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